

SDEV 220: PCRF Aid-Flow Inventory System – Final Project Documentation

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Date: May 9, 2026

I. Project Overview

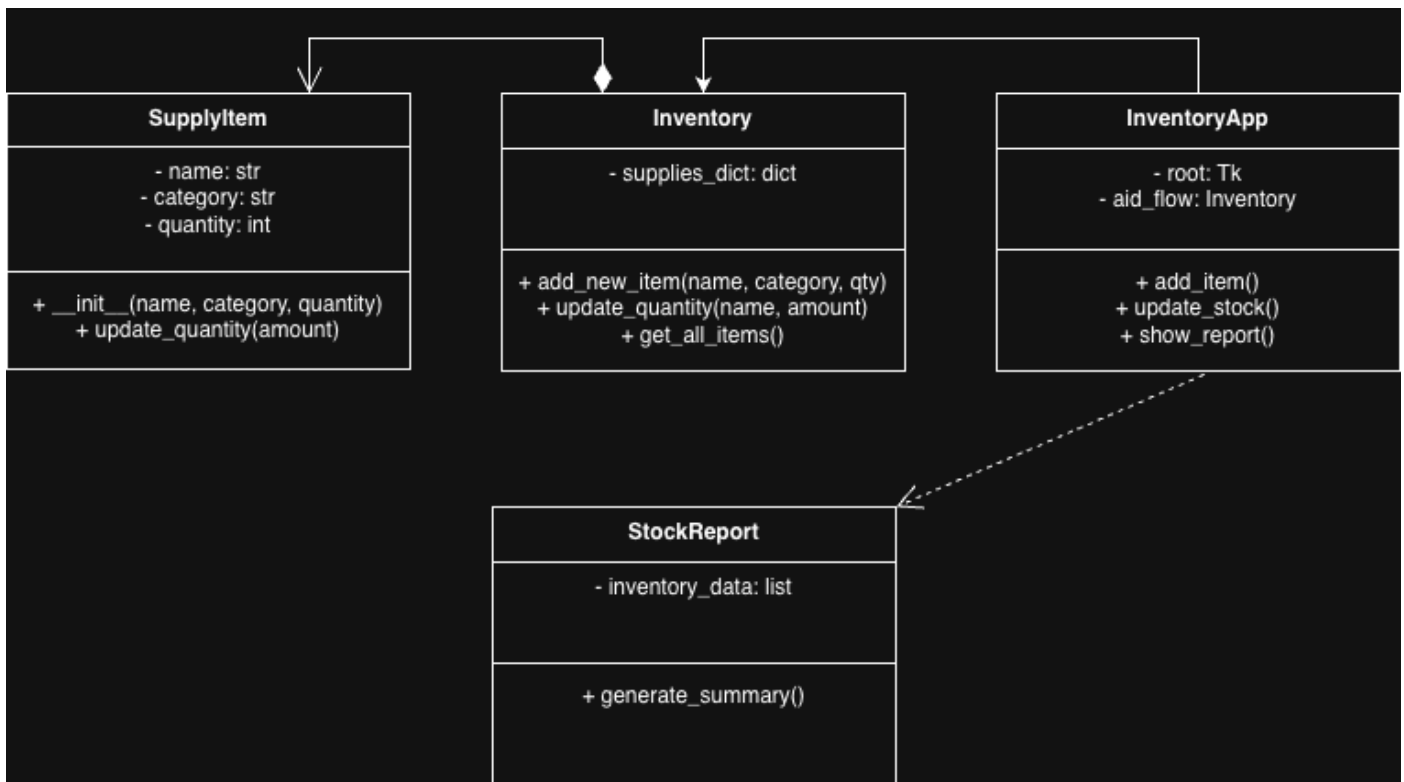
For our final project, the two of us developed the **PCRF Aid-Flow Inventory System**. We chose to design this for the Palestine Children's Relief Fund (PCRF) to assist in managing the intake & distribution of critical medical supplies. The goal was to create a user-friendly tool that makes sure humanitarian aid is tracked accurately in high-pressure environments/situations.

II. Technical Requirements Check

We verified that the system hit all technical requirements outlined in the rubric:

- **GUI:** Built using Tkinter library for a clean, dark-themed interface.
- **Classes:** System logic is split into three core classes: SupplyItem (data model), Inventory (manager), and StockReport (display logic).
- **Collections:** I used a Dictionary in the Inventory class for fast item lookups by name, and the StockReport class utilizes Lists to generate the final summary strings.

III. Class Diagram



IV. System Testing & Results

To authenticate the system's reliability in a simulated field environment, I took upon myself to perform a full logistics intake of five critical aid categories. This test confirmed that the backend dictionary collection correctly/efficiently manages unique keys and prevents data duplication.

- Successfully logged a diverse inventory ranging from PPE to specialized pediatric medicine.
- Performed a stock distribution test (updating 'Surgical Gloves' by -25) to ensure the system accurately calculates real-time stock levels.
- Verified the error-handling logic by attempting to enter non-numeric values; the system caught these inputs and alerted me without crashing.

PCRF Aid-Flow Inventory System

PCRF Aid-Flow Inventory System

Add New Supply

Name:

Category:

Quantity:

Add Item

Update Stock

Item Name:

Surgical Gloves

Amount (+/-):

-25

Stock Report

STOCK ALERT: Amoxicillin | Category: Antibiotics | Current Qty: 500

STOCK ALERT: Saline IV (500ml) | Category: Fluids | Current Qty: 300

STOCK ALERT: Sterile Gauze | Category: First Aid | Current Qty: 1200

STOCK ALERT: Pediatric Ibuprofen | Category: Pain Relief | Current Qty: 450

STOCK ALERT: Surgical Gloves | Category: PPE | Current Qty: 100

Generate Report

V. Individual Contributions

Danial Qasim (Project Lead / Backend): I handled the initial setup, wrote the core backend classes in models.py, and managed the GitHub and Trello environments. I also worked on the final integration to make sure the GUI and backend logic talked to each other correctly.

Aleeba Ghazanfar (Frontend & UI Design): Aleeba was a vital part of this project. She took full ownership of the frontend, designing an interface that is actually intuitive for a user. Beyond just writing the gui.py code, she was great at communicating throughout the week and was instrumental in testing the system to make sure the user experience felt right.

Simranjit Kaur (Absent): Despite reaching out multiple times on Discord and GitHub, Simranjit never responded or contributed to the project. Aleeba and I divided the work between the two of us to make sure we hit the deadline.